On the Correctness of Real Parameterization, the Modular Caveat, and Conditional Deduction under the Binary Normalization Hypothesis

Problem Statement

The following equalities are considered:

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

$$y^n = 2n\, l^n,$$

where the parameters

$$m, \; p, \; l$$

are not strictly required to be integers, but can be real numbers.

A further conditional step is also considered: if one accepts as a separate hypothesis, or an unproved axiom, the equality

$$o(n) = 2,$$

is the subsequent deduction

$$2^n = 2n$$

and, consequently,

$$n = 1 \quad \text{or} \quad n = 2$$

correct?

The new version of the formalization additionally takes into account the modular caveat. It is necessary in order to explicitly separate equalities over

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R}$$

from congruences modulo

$$\mathbb{F}_q.$$

In other words, three different levels are considered:

real level,

integer level,

modular level.

Real parameterization is used to reconstruct the algebraic scheme. The integer level is needed for the connection with Fermat's Last Theorem. The modular level is not specifically used as a proof method, but the new version separately explains why modular solutions to congruences do not constitute counterexamples to the integer formulation.

1. Correctness of the Equalities for x^n and z^n

The equalities

$$x^n = (m^n - p^n)^n$$

and

$$z^n = (m^n + p^n)^n$$

are correct provided that it is preliminarily set that

$$x = m^n - p^n,$$

$$z = m^n + p^n.$$

This is the standard raising of equal quantities to the same power.

In other words, if

$$x = m^n - p^n,$$

then automatically

$$x^n = (m^n - p^n)^n.$$

Similarly, if

$$z = m^n + p^n,$$

then automatically

$$z^n = (m^n + p^n)^n.$$

It is not required here for m and p to be integers. It is sufficient that the corresponding expressions are defined in the numerical domain under consideration, for example, in the domain of real numbers.

In Coq, this part corresponds to the lemma on raising parametric equalities to a power. The meaning of this lemma is that if the equalities

$$z = m^n + p^n, \qquad x = m^n - p^n,$$

are given, then after raising to a power, one automatically obtains

$$z^n = (m^n + p^n)^n, \qquad x^n = (m^n - p^n)^n.$$

Thus, there is no hidden mathematical assumption at this step: only the elementary property of equality is used.

2. Correctness of the Equality $y^n = 2n\, l^n$ for Real l

The equality

$$y^n = 2n\, l^n$$

is correct if l is understood as a real scale parameter.

After the binomial expansion of the expression

$$(m^n + p^n)^n - (m^n - p^n)^n$$

a sum over only the odd binomial terms is obtained:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2 \sum_{j \text{ odd}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Let us denote the inner sum by

$$S = \sum_{j \text{ odd}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Then

$$y^n = 2S.$$

If we now define the real parameter l by the condition

$$l^n = \frac{S}{n},$$

we obtain

$$S = n\, l^n.$$

Consequently,

$$y^n = 2S = 2n\, l^n.$$

Thus, the equality

$$y^n = 2n\, l^n$$

is correct as a real notation when l is defined as

$$l^n = \frac{1}{n} \sum_{j \text{ odd}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

At the same time, l is not required to be an integer.

If n is odd, the real root of S/n exists for any real value of S/n. If n is even, then for a real positive l, it is required to choose a situation where

$$\frac{S}{n} > 0.$$

It is exactly this positive case that is formalized in Coq through the positivity of the parameter l.

3. Important Difference from Integer Divisibility

If l is considered as a real number, there is no need to assert that the sum

$$S = \sum_{j \text{ odd}} \binom{n}{j} (m^n)^{n-j} (p^n)^j$$

is divisible by n in integers.

That is a different, stronger statement.

For real parameterization, it is sufficient to define

$$l^n = \frac{S}{n}.$$

Therefore, a potential issue with integer divisibility does not destroy the real scheme, provided it is explicitly stated that

$$l \in \mathbb{R}.$$

Moreover, the new Coq version specifically establishes an important warning: the statement that the odd binomial kernel is always divisible by n in integers is false in the general case. For instance, for

$$n = 6, \qquad m = 1, \qquad p = 1$$

the odd kernel takes the form

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32,$$

and the number 32 is not divisible by 6 in integers.

This does not break the real scheme because it does not assert the integer divisibility of the kernel by n. The real scheme uses the definition

$$l^n = \frac{S}{n},$$

rather than the assertion

$$n \mid S$$

in the ring of integers.

Consequently, one must strictly distinguish between:

$$l^n = \frac{S}{n}$$

as a real definition, and

$$n \mid S$$

as an integer arithmetic statement.

The former is permissible in the real model. The latter requires a separate proof and is generally incorrect.

4. Transition to the Scale Parameter o

If we introduce the scale parameter o via the equality

$$y = o\,l,$$

then for $l \neq 0$ we have

$$y^n = o^n l^n.$$

On the other hand, according to the previous equality

$$y^n = 2n\,l^n.$$

Consequently,

$$o^n l^n = 2n\,l^n.$$

If

$$l \neq 0,$$

one can cancel out l^n to obtain

$$o^n = 2n.$$

In other words,

$$o = (2n)^{1/n}$$

given the chosen positive real branch of the root.

In Coq, this corresponds to the definition

$$\text{covers\_with } o\ n,$$

which means

$$o^n = 2n.$$

More precisely, the Coq lemma on positive scaling states the following. If

$$n > 0,$$

$$l > 0,$$

$$y = o\, l,$$

and

$$y^n = 2n\, l^n,$$

then necessarily

$$o^n = 2n.$$

This is precisely the transition used in the article: from the real scaling equality to the cover condition

$$\text{covers\_with } o \ n.$$

5. Conditional Step $o(n) = 2$

Next, a separate binary normalization hypothesis is introduced:

$$o(n) = 2.$$

It is important to emphasize that this is exactly a hypothesis, not an automatically proven fact.

Upon accepting this hypothesis, from

$$o(n)^n = 2n$$

we obtain

$$2^n = 2n.$$

This is now a purely elementary exponential-linear equation.

It is equivalent to

$$n = 2^{n-1}.$$

Its positive integer solutions are:

$$n = 1$$

and

$$n = 2.$$

For all

$$n > 2$$

we have the strict inequality

$$2^n > 2n.$$

Consequently, under the condition

$$o(n) = 2$$

the exponents

$$n > 2$$

are excluded.

This particular conditional step is the core of the model. Coq strictly verifies that from

$$o(n) = 2$$

and

$$o(n)^n = 2n$$

it follows that

$$n \in \{1, 2\}.$$

But Coq does not unconditionally prove that

$$o(n) = 2$$

follows solely from the real parameterization. This condition remains a structural hypothesis of binary normalization.

6. The Modular Caveat

The new version of the article and the Coq file separately addresses the modular objection.

The present model does not construct a proof through modular arithmetic and does not assert the absence of solutions to the corresponding congruences in finite fields. The scheme under consideration applies to equalities over

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R},$$

and not to equations over the finite field

$$\mathbb{F}_q.$$

In particular, the integer equality

$$x^n + y^n = z^n$$

indeed implies the congruence modulo any q:

$$x^n + y^n \equiv z^n \pmod{q}.$$

However, the converse transition is generally incorrect: the congruence

$$x^n + y^n \equiv z^n \pmod{q}$$

does not imply the equality

$$x^n + y^n = z^n$$

in integers.

In other words, a modular congruence merely means that there exists some integer t for which

$$z^n = x^n + y^n + qt.$$

An integer solution to Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions for the congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

for

$$n > 2$$

is not a counterexample to Fermat's Last Theorem and does not refute the real parameterization used in this model.

If one specifically transitions to modulo q, the parameters must be set separately at the modular level. For example, instead of the real equalities

$$z = m^n + p^n, \qquad x = m^n - p^n$$

one should separately fix the residues

$$m^n \equiv A \pmod{q}, \qquad p^n \equiv B \pmod{q}.$$

After this, the corresponding modular relations take the form

$$z \equiv A + B \pmod{q}, \qquad x \equiv A - B \pmod{q}.$$

Thus, real parameterization and modular arithmetic belong to different levels of description. This article uses the first level: equalities over

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R},$$

rather than a separate theory of solutions over finite fields.

7. How the Modular Caveat is Reflected in Coq v6

In the sixth version of the Coq file, a special section

Section ModularRemark

has been added.

In it, the modular congruence relation is defined as follows:

$$a \equiv b \pmod{q}$$

means that there exists an integer t such that

$$a = b + qt.$$

In the Coq language, this is specified as

modular_congruent $q\ a\ b$.

Next, three key propositions are formalized.

First, if the equality of powers

$$x^n + y^n = z^n,$$

holds in integers, then the corresponding congruence modulo any q holds automatically:

$$z^n \equiv x^n + y^n \pmod{q}.$$

This corresponds to the lemma

integer_equality_implies_modular_power_congruence.

The meaning of the proof is simple: for a true equality, one can take

$$t = 0.$$

Second, Coq explicitly states that equality in integers corresponds to a zero modular witness:

$$t = 0.$$

This is reflected in the lemma

$$\text{integer\_equality\_is\_zero\_modular\_witness}.$$

Third, Coq verifies a specific example showing that a modular congruence is not the same as an integer equality:

$$3^3 \equiv 1^3 + 1^3 \pmod 5,$$

since

$$27 = 2 + 5 \cdot 5,$$

but at the same time

$$1^3 + 1^3 \neq 3^3$$

as an equality in integers, because

$$2 \neq 27.$$

This corresponds to the lemma

$$\text{modular\_congruence\_not\_integer\_equality}.$$

Thus, Coq v6 strictly confirms the methodological idea of the modular caveat:

$$\text{integer equality} \Longrightarrow \text{modular congruence},$$

but

$$\text{modular congruence} \not\Longrightarrow \text{integer equality}.$$

In addition, the structure

$$\text{ModularParameterResidues}$$

formalizes the idea that when transitioning to modulo q, one must separately specify the residues

$$m^n \equiv A \pmod q, \qquad p^n \equiv B \pmod q,$$

and only then separately specify

$$z \equiv A + B \pmod q, \qquad x \equiv A - B \pmod q.$$

This is fully consistent with the modular caveat: modular parameters are a separate level of description, not a direct replacement for real parameterization.

8. The Purpose of the Verification in Coq v6

It is exactly such a conditional scheme that is formalized in the sixth version of the Coq file.

It introduces a structure where the parameters

$$m,\ p,\ l,\ x,\ y,\ z$$

are treated as real numbers.

In this structure, the equalities

$$x^n = (m^n - p^n)^n,$$
$$z^n = (m^n + p^n)^n,$$
$$y^n = 2n\, l^n$$

are specified as data fields.

It is further proven that if additionally

$$y = o\, l$$

and

$$l > 0,$$

then from

$$y^n = 2n\, l^n$$

it follows that

$$o^n = 2n.$$

This is exactly what corresponds to the condition

$$\text{covers\_with}\ o\ n.$$

The sixth version of the Coq file also separates the real level from the integer one. For real parameters, a block is used where

$$m, p, l, x, y, z \in \mathbb{R}.$$

For the integer level, a separate structure is introduced where the parameters lie in

$$\mathbb{Z}.$$

At the integer level, additional arithmetic conditions appear, for example, parity conditions:

$$z + x \text{ is even}, \qquad z - x \text{ is even}.$$

This aligns with the mathematical idea of the article: the real reconstruction can be formally correct, but if integer parameters are additionally required, additional restrictions arise.

9. Consistency of the Fourth, Fifth, and Sixth Versions

The fourth version of the Coq file verifies the main conditional scheme:

$$\text{if a hypothetical solution implies } o^n = 2n,$$

and if

$$o = 2,$$

is then assumed, then only

$$n = 1 \quad \text{or} \quad n = 2.$$

is possible.

The fifth version expands the fourth.

It adds an explicit real structure for the parameters

$$m,\ p,\ l,\ x,\ y,\ z$$

and formally shows the transition

$$y = o\,l, \qquad y^n = 2n\,l^n \quad \Longrightarrow \quad o^n = 2n.$$

The sixth version expands the fifth.

It adds the modular caveat at the Coq level and formally shows that:

$$\text{equality in integers}$$

and

$$\text{congruence modulo}$$

are different logical objects.

Therefore, the fourth, fifth, and sixth versions are consistent with each other.

The fourth version verifies the final conditional exponential-linear part.

The fifth version adds the real level of parameterization.

The sixth version adds the modular level and shows that modular congruences are not counterexamples to the integer or real formulation.

Thus, the sixth version does not contradict the fourth and fifth, but clarifies and expands them by adding an explicit separation:

$$\text{real parameterization,}$$

$$\text{integer specialization,}$$

$$\text{modular caveat.}$$

10. What Exactly Coq v6 Proves

Coq v6 proves the following propositions.

10.1. Correctness of Elementary Parametric Equalities

If

$$z = m^n + p^n, \qquad x = m^n - p^n,$$

then

$$z + x = 2m^n, \qquad z - x = 2p^n.$$

In the real case, these are ordinary equalities over

$$\mathbb{R}.$$

In the integer case, additional parity conditions follow from them.

10.2. Correctness of the Transition from $y = o\,l$ to $o^n = 2n$

If

$$y = o\,l,$$

$$l > 0,$$

and

$$y^n = 2n\,l^n,$$

then

$$o^n = 2n.$$

This is the main real transition of the model.

10.3. Correctness of the Final Conditional Exponential Part

If one assumes

$$o = 2,$$

then from

$$o^n = 2n$$

it follows that

$$2^n = 2n.$$

Coq proves that in positive natural numbers this is possible only when

$$n = 1 \quad \text{or} \quad n = 2.$$

Consequently, for

$$n > 2$$

a contradiction arises.

10.4. Correctness of the Modular Caveat

Coq v6 proves that a modular congruence follows from an integer equality, but a modular congruence by itself is not an integer equality.

Formally:

$$x^n + y^n = z^n \quad \Longrightarrow \quad x^n + y^n \equiv z^n \pmod{q}.$$

But the converse is generally incorrect:

$$x^n + y^n \equiv z^n \pmod{q} \quad \not\Longrightarrow \quad x^n + y^n = z^n.$$

This resolves the methodological objection that modular examples allegedly automatically refute the real parameterization.

11. What Coq v6 Does Not Prove

For the sake of fairness, it is important to explicitly state what Coq v6 does not prove.

11.1. Coq v6 Does Not Unconditionally Prove $o(n) = 2$

The equality

$$o(n) = 2$$

remains the binary normalization hypothesis.

Coq proves that if this hypothesis is accepted, the subsequent deduction is correct.

But Coq does not prove that

$$o(n) = 2$$

follows solely from the real parameterization.

11.2. Coq v6 Does Not Prove the Full Theory of Finite Fields

The modular section does not construct a full-fledged theory of the field

$$\mathbb{F}_q.$$

It does not require q to be prime, does not introduce the multiplicative group of the field, and does not analyze invertible elements.

The modular section fulfills a more modest and precise task: it formalizes the distinction between equality in integers and congruence modulo.

11.3. Coq v6 Does Not Prove Unconditional FLT

The file proves the conditional scheme:

if a hypothetical solution for

$$n > 2$$

implies

$$\text{covers\_with } 2\ n,$$

then a contradiction arises.

That is, it does not prove Fermat's Last Theorem unconditionally, but rather a conditional theorem of the form:

$$(x^n + y^n = z^n \Longrightarrow 2^n = 2n) \Longrightarrow \text{no solutions for } n > 2.$$

This is exactly the status that should be fairly stated in the article.

12. General Interpretation of the New Logic

The new logic of the article can be formulated as follows.

First, Fermat's equation

$$x^n + y^n = z^n.$$

is considered.

Then it is rewritten as

$$y^n = z^n - x^n.$$

Next, the parameterization

$$z = m^n + p^n, \qquad x = m^n - p^n.$$

is introduced.

After that, we obtain

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

The binomial expansion yields an outer binary coefficient of

$$2.$$

Then, the real scale kernel

$$l$$

and the scale parameter

$$o.$$

are introduced.

Given the conditions

$$y = o\,l$$

and

$$y^n = 2n\,l^n$$

it follows that

$$o^n = 2n.$$

Next, binary normalization

$$o = 2.$$

is adopted as a separate structural hypothesis.

Then

$$2^n = 2n.$$

This equation only has the solutions

$$n = 1 \quad \text{and} \quad n = 2.$$

Consequently, for

$$n > 2$$

a hypothetical solution leads to a contradiction.

The modular caveat states that the congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

may have solutions for

$$n > 2,$$

but this is not an objection to the scheme, because such congruences are not integer equalities

$$x^n + y^n = z^n.$$

Final Conclusion. *Under the understanding that*

$$m, p, l \in \mathbb{R}$$

the equalities

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

$$y^n = 2n\, l^n$$

are correct as a real formal scheme.

The sixth version of the Coq file consistently formalizes this scheme via real parameters, separately fixes the integer specialization, and additionally introduces the modular caveat.

If one then separately accepts the binary normalization hypothesis

$$o(n) = 2,$$

the subsequent deduction

$$2^n = 2n$$

and the restriction

$$n \in \{1, 2\}$$

are correct and machine-verified.

Consequently, under the specified formulation, the underlying conditional logic is consistent: the weak point lies neither in the real parameterization, nor in the equality

$$y^n = 2n\, l^n,$$

nor in the modular examples, but exclusively in the status of the hypothesis

$$o(n) = 2.$$

Modular congruences for

$$n > 2$$

may exist, but they belong to a different level of the problem and are not integer solutions to Fermat's Last Theorem. This is precisely what is now explicitly reflected in both the article's text and the Coq file v6.